

Mistral Solutions

Hardware Role

Round 1:

1. Diode problems – find the output both zener and PN
2. Clipper, clamper
3. Inverting, non-inverting, voltage follower problems
4. Hartley, RC, amplifier oscillator frequency equation
5. Barkhausen criteria sums
6. Resonance formula sums
7. Window, zero crossing detector
8. Schmitt trigger formula apply sums
9. Op-amp one circuit output to another circuit
10. Monostable, astable time period, frequency calculation
11. ADC – see all ADC comparison like resolution, speed, conversion time, which is fastest, conversion time calculation, comparators needed, reference voltage calculation, clock cycle
12. Comparator problems.

Round 2 and Round 3:

second and third round all the questions were basics but question were covered from wide range of topics also explain every question with an example they will also ask u to draw circuits for op-amp based questions topics were from circuits and networks, electronic circuits, dsp, mpmc(very basic), adc and awp(just imp topics), lic(See all imp circuits and formulas), digital electronics (see all topics)

1. Circuits and Networks

- What is resistor and working, principles
- Capacitor working, principles
- Inductor working , principles
- Norton
- Thevenin
- Superposition
- Maximum power transfer
- KVL and KCL
- Resonance curve
- Curve at $f = 1/2\pi RC$

- What is cut-off frequency
- Why -3 dB point is used

2. Electronic Circuits

- PN junction diode
- BJT types and diagram
- Active vs passive devices and example
- JFET vs MOSFET working and pin diagram
- Amplifiers and oscillators difference
- Darlington amplifier

3. Linear Integrated Circuits (LIC)

- High-pass filter / differentiator op-amp circuit
- Low-pass / integrator op-amp circuit
- Monostable vs astable
- Inverting, non-inverting op-amp, all diagrams and V_o formulas
- What is CMRR
- What is op-amp
- What is comparator
- Schmitt trigger
- 555 timer

4. Digital Electronics

- K-map use
- Decoder
- Encoder
- MUX
- DEMUX
- All flip-flops
- Latches
- RAM
- ROM
- Shift registers
- Synchronous & asynchronous counters
- 'n' bit counter using FF
- See one FF to another FF conversion

5. Digital Signal Processing (DSP)

- FFT vs DFT
- Fourier transform
- FIR
- IIR
- What is sampling
- Why quantization is needed

6. ADC and DAC

- ADC conversion process explain
- All ADC types
- DAC types

7. Communication Systems

- What is the use of antenna
- Analog communication
- Digital communication
- SSB
- DSB
- Gain
- Directivity
- Bandwidth

Round 4:

In this round, they asked about all the projects present on the resume. They also asked why I used each component and the specifications of those components. They asked about all the skills and training mentioned in the resume.

Be prepared about everything you put on your resume. Prepare how to explain your projects, and refer to the specifications and features of all the tools and technologies you used in your projects, like if you used any sensor see which protocol used for sensor communication and see sensor operating voltage type analog or digital as well as the previous companies you attended and why you were not selected by those companies.

Round 5:

Basic HR questions were asked, like:

- Why should we hire you?
- Questions about your family.
- Questions about your parents' occupation.
- What do you know about Mistral?
- Why do you want to join Mistral?
- What do you think about Mistral?
- Are you okay with relocating to Bangalore?